

NAT Configuration — HomeSOClab

NAT is configured under **Firewall** → **NAT** in pfSense/OPNsense.

1. Outbound NAT (Source NAT / masquerade)

Switch **Firewall** → **NAT** → **Outbound** to **Hybrid Outbound NAT** so we can write explicit rules while keeping pfSense's automatic ones for the firewall itself.

#	Interface	Source	Translation	Description
1	WAN	10.10.10.0/24 (MGMT)	WAN address	Management in- ternet egress
2	WAN	10.10.20.0/24 (ATTACKER)	WAN address	Attacker tooling egress

Deliberately NOT NATed: the **Victim (30)** and **Monitoring (99)** subnets have no outbound NAT entry. Even if a firewall rule were misconfigured, the absence of a NAT translation means their traffic cannot be masqueraded onto the WAN — a defense-in-depth “belt and suspenders” so victims/sensors stay dark.

Why hybrid, not manual-only? Manual mode forces you to recreate every system NAT (e.g., for the firewall's own updates). Hybrid keeps those working while giving us explicit control over the lab zones.

2. Port Forwards (inbound NAT) — optional

Phase 1 needs **no inbound port forwards** (no lab service is exposed to the home LAN/internet — by design). If, in a later phase, you want to reach the SIEM dashboard from the home LAN, prefer a controlled jump path instead of a raw port forward. A documented example (disabled by default) if you must:

IF	Proto	Ext port	Target	Int port	Note
WAN	TCP	8443	10.10.10.10	443	SIEM UI — disabled ; enable only if required

Security note: exposing lab management to the home network widens the blast radius. Keep inbound NAT empty unless you have a specific, audited need, and always pair it with a strict source restriction (your admin host only).

3. NAT reflection

Leave **NAT reflection = disabled** (default). The lab has no internal clients that need to reach internal services via the WAN address.